

Title

Smart Traffic Management Using Artificial Intelligence

Abstract

Traffic congestion is a major problem in urban areas. This paper proposes an Artificial Intelligence-based traffic management system that uses computer vision and machine learning to optimize traffic signal timing. Cameras capture real-time traffic data, and a deep learning model detects vehicle density. Based on the predicted traffic flow, the system dynamically adjusts signal durations, reducing congestion and travel time. Experimental results show that the proposed system improves traffic efficiency by 25% compared to traditional fixed-time traffic signals.

Introduction

Rapid urbanization has increased the number of vehicles on roads, leading to traffic congestion, pollution, and delays. Conventional traffic signal systems operate on fixed schedules and cannot adapt to changing traffic conditions. Artificial Intelligence enables real-time decision-making by analyzing traffic patterns and predicting congestion.

Methodology

The proposed system consists of traffic cameras for image collection, a CNN to detect vehicle density, and a reinforcement learning algorithm that adjusts signal timing.

Results

The AI-based system reduced waiting time by 25%, fuel consumption by 18%, and improved traffic flow.

Conclusion

Artificial Intelligence can significantly improve urban traffic management by making traffic signals adaptive and intelligent.